

Robot catch temporal and tracking annotation

EVALUATION REPORT

2026-09-07

Evaluation scope:

meta/muse-spark-1.3

Prepared by:

VVDex Forge — generated directly from sealed run evidence

fr-20260907-01cca689 · vvdex.annotation.robot-video-1 v0.2.0 · Independent model evaluation report

About this report

About VVDex Forge

VVDex Forge builds and certifies evaluation environments for AI agents. Every exam is sealed before any model sees it: the reference solution must pass, an empty submission must fail, the grader must reject near-misses, and the hidden tests never enter the agent's tree. Reports and the public catalog live at **vvdexops.com**.

Report integrity

This report is generated from run evidence; its digests verify authenticity. The sole test of a copy is the three-line procedure in Verification — a copy whose digests do not reproduce is not this report.

Scope

One exam · 2 rollout(s) · 24 tool steps and 2280s wall clock per rollout, stated in full in Evaluation configuration.

Boundary

Not a leaderboard. Not a general capability score. A post-seal benchmark: this run stamped no certification gate and rewrote nothing — not the exam, not the reference solution, not the grader, not the certified evidence.

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Followed by: companion evidence report reference · Attestation

01 Annotation evidence

Modality: video. **Rubric:** 1.1.

A robot-catching video is reviewed for temporal events, frame-level object boxes, track identity and occlusion.

Submission ae21e559-693d-4e31-a388-65074b74ce97

Score	Items	Valid / invalid	Types
0.6632	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1
boxIoU	0.5729	boxPrecision	1
boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1

classificationRecall	1	eventF1	0.5
eventPrecision	0.5	eventRecall	0.5
eventTemporalIoU	0.4667	matchedCount	6
mismatchCount	5	segmentF1	1
segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	0.9	Track consistency on matched boxes	1

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement reviewNot measured

Submission 7a9d5c84-a8e4-4e4e-8284-3632138829dd

Score	Items	Valid / invalid	Types
0.8703	6	6 / 0	box, classification, event, segment

Measured metrics

annotationCount	6	boxF1	1
boxIoU	0.7359	boxPrecision	1
boxRecall	1	classificationAccuracy	1
classificationF1	1	classificationPrecision	1
classificationRecall	1	eventF1	1
eventPrecision	1	eventRecall	1

eventTemporalIoU	0.875	matchedCount	6
mismatchCount	3	segmentF1	1
segmentPrecision	1	segmentRecall	1
segmentTemporalIoU	1	Track consistency on matched boxes	1

Track consistency depends on successful box matching. A zero can reflect inaccurate boxes even when track IDs are correct.

Validation error codes

None recorded

Disagreement

Agreement reviewNot measured

Measured grader aggregates only. Missing metrics or disagreement are not measured, never zero. Reference answers and item-level labels are excluded. Certification and the evidence digest are recorded in this report's verification sections.

02 Executive summary

MODELS TESTED

1

CERTIFIED PASSES

1

of 2 graded rollouts

SUBMITTED

2 of 2

graded rollouts that called submit

NOT GRADED — LANE ERRORS

0

of 2 attempts

Purpose

This report presents a controlled evaluation of 1 model lane(s) against the certified exam `vvdex.annotation.robot-video-1` (v0.2.0, family annotation). Every lane received the identical frozen workspace, tool contract and limits; grading ran afterwards, in isolation, against hidden tests no lane ever saw. The purpose is to state, with reproducible evidence, what each lane did and where it stopped.

Outcome

1 of 2 graded rollouts passed the independent hidden tests and were certified

1 of 2 graded rollouts passed. `meta/muse-spark-1.3` completed the applicable annotation workflow and were accepted by the independent annotation grader. **Low-N:** 2 rollout(s) is below the 10-rollout floor for reading a rate as a rate — read the per-rollout outcomes, not the percentages. The most common way to fail was `submitted_failed` — submitted annotations rejected by the independent grader.

Key findings

F-01 `meta/muse-spark-1.3` passed the independent hidden tests

F-02 `submitted_failed` × 1 (`meta/muse-spark-1.3`)

F-03 Statistical floor — 2 rollout(s) is below the 10-rollout rate floor

Each finding is stated in full, with evidence, in the Findings section; recommendations for acting on them follow it. Elapsed wall clock for the whole engagement: 510.7s.

03 Exam definition

In scope. The ability of each configured model lane to complete this one certified task end to end — inspect the asset, annotate against the declared rubric, validate, submit, and be accepted by the independent annotation grader — within 24 tool steps and 2280s of wall clock, at 2 rollout(s) per lane. Behavioural measurements along the way (tool discipline, grounding, overclaim) are in scope because they are read from the recorded trace, not judged by another model.

Out of scope. General model quality, performance on other task families, prompt-tuning on the lane's behalf, and any comparison this run's sample size cannot support. A lane's API failure is reported as infrastructure, never as model capability.

The instrument. Exam `vvdex.annotation.robot-video-1` was certified before this run: its reference solution passes, an empty submission fails, its grader distinguishes near-

misses, and its sealed evidence is unchanged by this evaluation (see Certification evidence).

What was measured

EXAM

**vvdex.annotation.robot
-video-1**

VERSION

0.2.0

FAMILY

annotation

ROLLOUTS

2 (2 per model)

Where the defect came from

This exam has no upstream repository to credit: the defect, the reference solution and the hidden suite are VVDex's own.

04 Certification evidence

Why this exam is trustworthy. Every pillar below was established before any model saw the exam, loaded from the sealed evidence matching this run's exam fingerprint, and unchanged by this evaluation.

FROZEN BASELINE

FAIL

expected — an empty submission over 2 run(s)
must not pass

→

GOLD / REFERENCE

PASS

expected — the reference solution over 2 run(s)
must pass

GRADER CONTROLS

19 / 19

The grader accepted the reference and rejected every near-miss.

ATTACK PROBES

9 / 9 blocked

0 trivial exploit(s) found.

SEALED EVIDENCE

verified

Sealed 2026-09-06 over 13 artifact(s).

RUNTIME

sha256:679b36225a1f
83238efba8c71b9cde
3c24caaeacc9b682ae
92819cb55eec328f

network policy: deny

CERTIFICATION BUNDLE DIGEST

0a0a51360cde69737c9b5caef895b8b7c2a5e2d66d
a89617d8776ac724b019b9

EXAM FINGERPRINT

85e026007af17db692614084...

Full value in Appendix B; the contract digest this run was executed against.

05 Model outcomes

Denominators. Attempts requested: 2. Graded (evaluable) rollouts: 2. Not graded — lane errors: 0. A lane error is a rollout the API or the runtime ended before the hidden grader ran: it is listed, excluded from every rate and pass@k, and never silently dropped. Passes are certified passes: hidden grader exit 0 with integrity verified.

MODEL	ATTEMPTS	GRADED	PASSED	MODEL FAILS	LANE ERRORS	SUBMITTED	MEAN TIME	TYPICAL DEEPEST STAGE
meta/muse-spark-1.3 meta/muse-spark-1.3	2	2	1 / 2	1	0	2 / 2	255.1s	Passed

Every rollout, with its own deepest stage and grade, is in Appendix A.

Success rate, confidence and pass@k

Every rate on this page is a measurement of a finite number of rollouts, so every rate carries the interval that measurement actually supports. n is the evaluable count — attempts minus rollouts the lane ended before grading; those are listed under "not graded" and sit in no rate, interval or pass@k.

MODEL	ATTEMPTS	N	NOT GRADED	PASSES	RATE	95% INTERVAL	SEEDS	PASS@1	PASS@2
meta/muse-spark-1.3	2	2	0	1	50.0%	[9%, 91%]	0, 1	50.0%	100.0%

Interval and pass@k methods are stated in full in Appendix D. pass@k is shown for k = 1, 2; the sealed record carries every k up to n.

Model comparison

One model ran in this record, so there is nothing to compare it with. A comparison table across models appears here when a run covers more than one.

06 Stage funnel

Recorded annotation actions and independent grader verdicts. Missing capabilities are N/A; each stage is measured independently.

STAGE	OPENROUTER:META/MUSE-SPARK-1.3
Inspect	2 / 2
Annotate	2 / 2
Validate	2 / 2
Submit	2 / 2
Graded	2 / 2
Passed	1 / 2

Deepest stage reached

MODEL	DEEPEST (ANY ATTEMPT)	TYPICAL (MOST ATTEMPTS)	ANNOTATION QA PASSES	NOT GRADED
openrouter:meta/muse-spark-1.3	Passed	Passed	1 / 2	0

The matrix counts evaluable annotation attempts; lane errors are shown separately. Successful recorded actions establish each stage independently.

07 Findings

Every finding below is derived from the recorded data of this run — none is an opinion. Severity: PASS a lane succeeded · ATTENTION a capability gap · LANE / INFRA infrastructure, not capability · NOTE a property of the measurement.

F-01

PASS

meta/muse-spark-1.3 passed the independent hidden tests

Severity: PASS

Evidence: Model outcomes · Stage funnel · Appendix A

1 of 2 graded rollouts completed the applicable annotation workflow and were accepted by the independent annotation grader. This demonstrates the task is solvable under the published limits.

F-01 · derived from the recorded run data

F-02

ATTENTION

submitted_failed × 1 (meta/muse-spark-1.3)

Severity: ATTENTION

Evidence: Appendix A · Appendix C

Submitted annotations rejected by the independent grader.

F-02 · derived from the recorded run data

F-03

NOTE

Statistical floor — 2 rollout(s) is below the 10-rollout rate floor

Severity: NOTE

Evidence: Appendix D

The counts are exact; the percentages are anecdotes with decimal points. Per-rollout outcomes, not rates, are the reliable reading of this run.

F-03 · derived from the recorded run data

08 Recommendations

One recommendation per observed failure mode, addressed to the lane it affects. Each traces back to a finding above and to the raw trace in Appendix A.

R-01 PROBLEM

Submitted annotations rejected by the independent grader.

AFFECTED LANES

meta/muse-spark-1.3

WHAT WE WOULD LOOK AT FIRST

Review the measured annotation metrics and validator error codes against the declared rubric.

R-02 For anyone reading the percentages

Commission a run at $n \geq 10$ rollouts per lane before treating any rate here as a rate; this run's per-rollout outcomes are exact, its percentages are not yet stable.

09 Verification

RUN ID

live-20260907T090941Z

EXAM FINGERPRINT

**85e026007af17db692614084f12ff40c5bbf2348bc
fda0a01deb95aad8928f4c**

The contract digest this run was executed against.

Tamper-evidence

Three digests, and how to check each one. If any of them does not reproduce, this page is not the page that was generated for this run.

THIS REPORT

**aa0fa7d50e74581b6037e8f4c8
704f849b95f519fdb965426957
3daa94dfd015**

sha256 over the report's UTF-8 bytes with every occurrence of the report digest itself replaced by 64 '0' characters

EVAL RECORD

**4683299eba4f1559ac3de97ca6
8d2537f8c27b61a1a313339c70
a6f31980fead**

sha256 of fr-20260907-01cca689.json as written beside this file.

RUN EVIDENCE BUNDLE

**01cca6890a7e0b3a93c94256cc
d921c27f4258ef23f0b293747b
3c6cbcc1f605**

The digest the run's own bundle computed over itself.

CERTIFIED EVIDENCE SEAL

**0a0a51360cde69737c9b5caef8
95b8b7c2a5e2d66da89617d877
6ac724b019b9**

The exam's sealed evidence, established before this run.

Measurement history

Previous run on this exam: fr-20260907-023b5cdb. Model progress over time on this exam is the difference between that record and this one — same frozen tree, same hidden grader, same limits. Anything else that moved between them moved outside this exam.

Verifying it, in three lines

1. `vvdex-env verify-report --report fr-20260907-01cca689.html`
2. or by hand: replace every occurrence of the report digest above with 64 zeros, then sha256 the file — it must equal that digest
3. `shasum -a 256 fr-20260907-01cca689.json` must equal the eval-record digest, and the bundle digest must equal `bundleSha256` in the run's bundle file

The full artifact-by-artifact digest table, and the reproduction protocol, are in Appendix B.

This is the executive report. The full verifiable evidence report for the same run — every rollout, artifact digests, redacted transcripts and methods — is fr-20260907-01cca689.pdf, generated from exactly the same sealed record.

Attestation

This report was generated end to end by VVDex Forge from the sealed evidence of the evaluation run described above. Measurement values were populated from recorded artifacts, not manually entered. The digests in Verification let any holder of this file confirm that the report, the eval record and the run evidence are the ones this run produced — without contacting VVDex.

VVDex Forge · vvdexops.com	2026-09-07	vvdex-env 0.1.0
Issuer — certified evaluation environments	Date of issue	Engine · verifiers pin c521e4146026

No upstream repository: this exam is VVDex’s own.

Post-seal benchmark. It does not stamp any certifying gate and did not rewrite the exam, the gold solution, the grader or the certified evidence.

— END OF REPORT · fr-20260907-01cca689 —